

★ WEEK 2 · PART 3 of 3

Spurious Regression & Cointegration

The Trap, and the Cure

Why unrelated series fake relationships — and how to find the real ones

Now we deliver the payoff of the 1974 episode: the exact mechanism of spurious regression, and cointegration — the elegant idea that separates a genuine long-run link from a statistical mirage.

3

PART 3 · COINTEGRATION

From the Trap to the Cure

The spurious-regression mechanism in full, then cointegration, the error-correction model, and the application that made both famous — pairs trading.

The Episode's Unfinished Business

We named the trap in Part 1 — now we open it up

1

Part 1 warned us; Part 2 armed us

We can now detect a unit root with ADF and KPSS. That is exactly the tool we need to understand — and defuse — spurious regression.

2

The trap: two $I(1)$ series can fake a relationship

Regressing one trending series on another unrelated one yields a high R^2 and a huge t-statistic that mean nothing at all.

3

The cure has a name: cointegration

Sometimes two $I(1)$ series really do move together. Cointegration is how we tell a real long-run link from a mirage — and trade it.

Part 3: first the mechanism of the mirage, then the mathematics of the real thing.

The Spurious Regression Trap, Formally

Why the residual is the tell

$$y_t = \alpha + \beta x_t + u_t, \quad x_t, y_t \sim I(1) \text{ independent} \Rightarrow u_t \sim I(1)$$

What goes wrong

- Regress two independent $I(1)$ series and the residual is itself $I(1)$.
- A non-stationary residual violates the regression's core assumption.
- So every standard error and t-statistic is invalid.

The diagnostic

- The giveaway is a non-stationary residual, not a low R^2 .
- In fact R^2 is often deceptively high.
- Always test the residual before you trust the regression.

Why R^2 and the t-Statistic Explode

The usual asymptotics simply fail

as $T \rightarrow \infty$: $|t_{\hat{\beta}}| \rightarrow \infty$, $R^2 \not\rightarrow 0$ (the t-test never settles)

The broken machinery

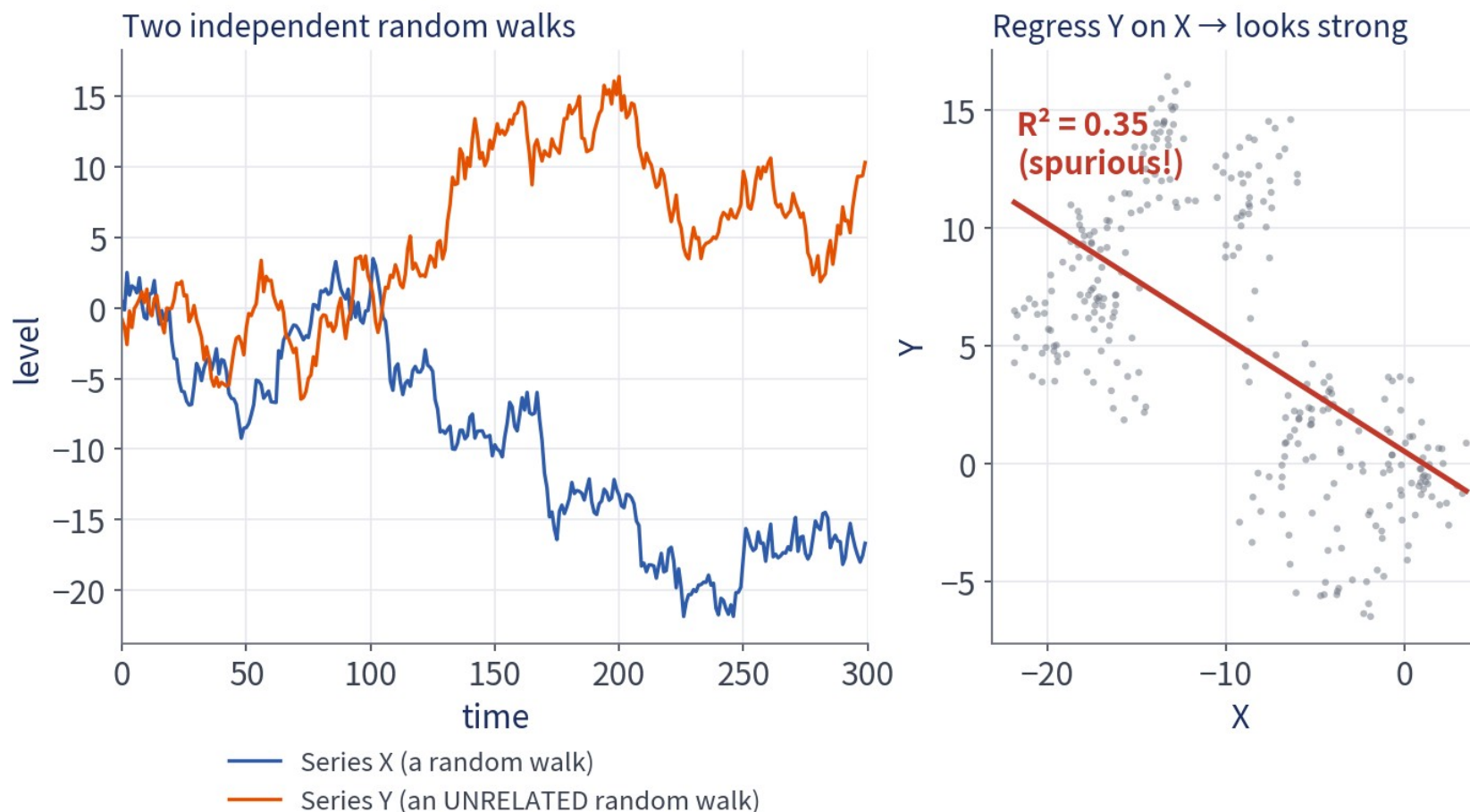
- For stationary data, the t-statistic settles to a fixed distribution.
- For two $I(1)$ series, it instead diverges as the sample grows.
- More data makes the illusion look STRONGER, not weaker.

The intuition

- Both series wander; by chance they drift in loose lockstep.
- The regression reads that shared wandering as 'correlation'.
- But there is no mechanism linking them — it is an artifact.

Two Unrelated Random Walks Fake a Relationship

Left: two independent random walks. Right: regress one on the other and you get a high R^2 and a confident-looking line — from series with no connection at all.

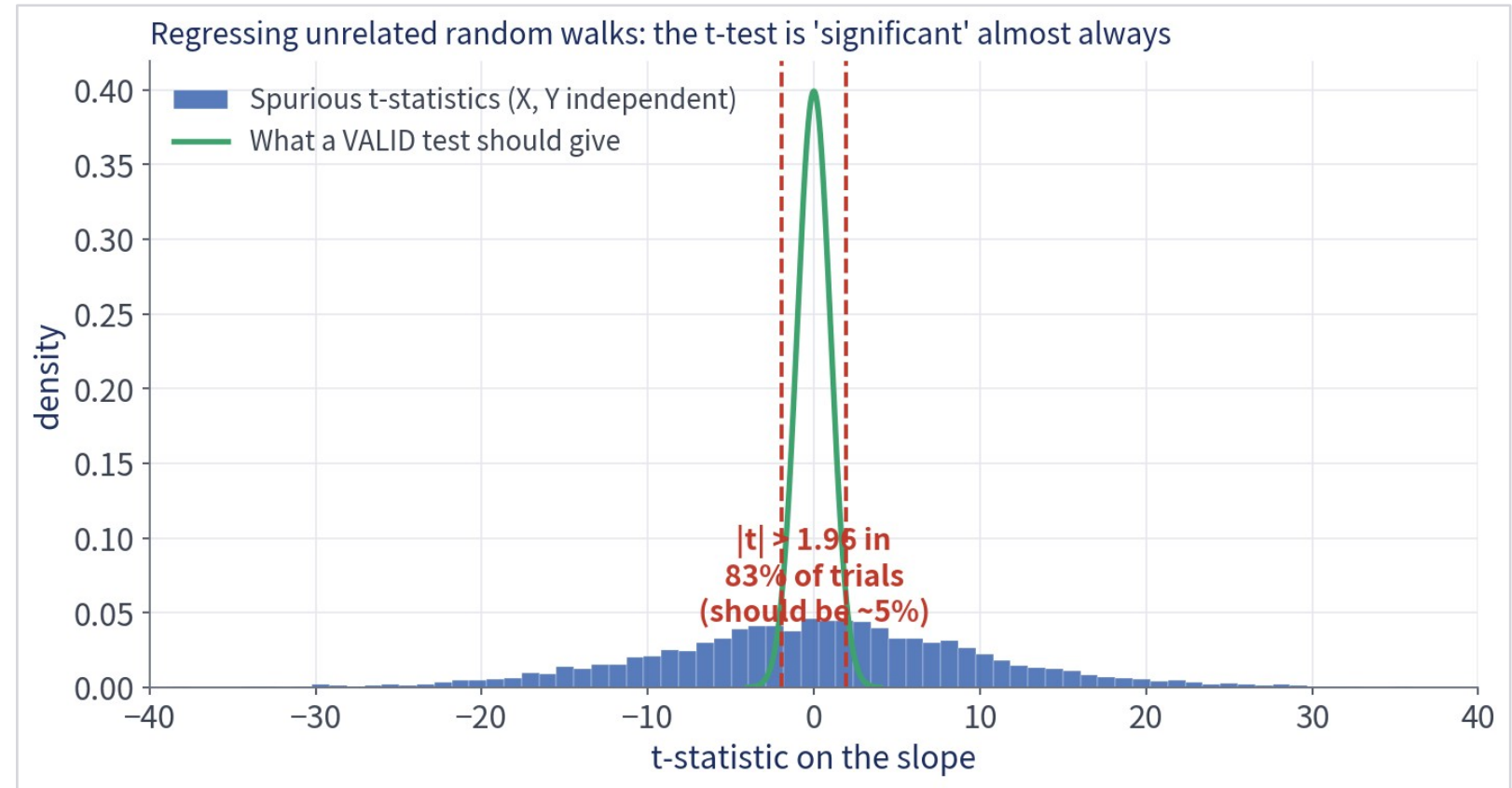


How Often the t-Test Lies

Repeat the experiment thousands of times: the t-test screams 'significant' almost every time.

What to notice

- Blue = spurious t's.
- They spread past ± 2 .
- Green = a valid test.
- $|t| > 2$ should be $\sim 5\%$.
- Here: the majority.



Granger and Newbold's Warning Sign

A quick check you can run on any regression

Warning sign: $R^2 > DW$ (Durbin–Watson) $\Rightarrow$ suspect spurious

How to read it

- The Durbin-Watson statistic detects residual autocorrelation.
- A very low DW alongside a high R^2 is a red flag.
- R^2 exceeding DW is the classic spurious-regression signature.

What to do next

- Do not trust the coefficients — investigate the residual.
- Test whether the residual is stationary ($I(0)$) or not.
- If it wanders, you are looking at a spurious regression.

How to Avoid Spurious Regression

Four habits that keep you honest

Do this	Instead of	Why it matters
Test each series for $I(1)$ first	regressing raw levels blindly	levels may share only a trend
Check cointegration before using levels	trusting a high R^2	R^2 is meaningless for $I(1)$ levels
Model differences or an ECM	modeling raw levels	differences/spread are stationary
Inspect the residual for $I(0)$	reading t-stats at face value	non-stationary residuals void inference

The single unifying habit: never regress two levels without checking what their residual does.

The Naive Fix: Regress Differences

Correct, but it comes at a cost

Differencing both series makes them stationary, so the regression becomes valid again. This is the safe, standard fix — but it quietly discards something valuable.

Why it works

- Differences of $I(1)$ series are $I(0)$ — stationary.
- On stationary data, t-tests and R^2 behave normally.
- For most purposes, model returns, not price levels.

What it loses

- Differencing removes the LONG-RUN relationship between levels.
- You can no longer say 'these two prices move together'.
- Sometimes that long-run link is exactly what you want →

But Differencing Throws Away the Long Run

When the levels genuinely belong together

Two prices can wander individually yet stay tethered — an arbitrage force pulls them back whenever they drift apart. Differencing erases that tether entirely.

The everyday example

- Two share classes of the same company.
- A stock and its close industry peer.
- Spot and futures on the same asset.

What we want instead

- A method that keeps the levels but stays valid.
- One that captures the tether, not just the wiggles.
- That method is cointegration.

Cointegration: A Real Long-Run Link

Two wanderers, one stationary combination

Two series are cointegrated if each is individually non-stationary, yet a particular linear combination of them IS stationary. That combination is the tether — and it mean-reverts.

The picture

- Each series alone is $I(1)$ — it wanders with no fixed mean.
- But their spread is $I(0)$ — it reverts to a stable level.
- The spread is what you actually model and trade.

Why it is deep

- It distinguishes a genuine equilibrium from a coincidence.
- Granger and Engle won the 2003 Nobel for formalizing it.
- It rescues the levels that differencing would have thrown away.

Cointegration, Formally

One equation captures the whole idea

$$X_t, Y_t \sim I(1), \quad \exists \beta : \quad Z_t = Y_t - \beta X_t \sim I(0) \quad (\text{cointegration})$$

Reading it

- X and Y are each $I(1)$ — non-stationary on their own.
- There exists a β making $Z = Y - \beta X$ stationary, $I(0)$.
- β is the cointegrating coefficient; Z is the spread.

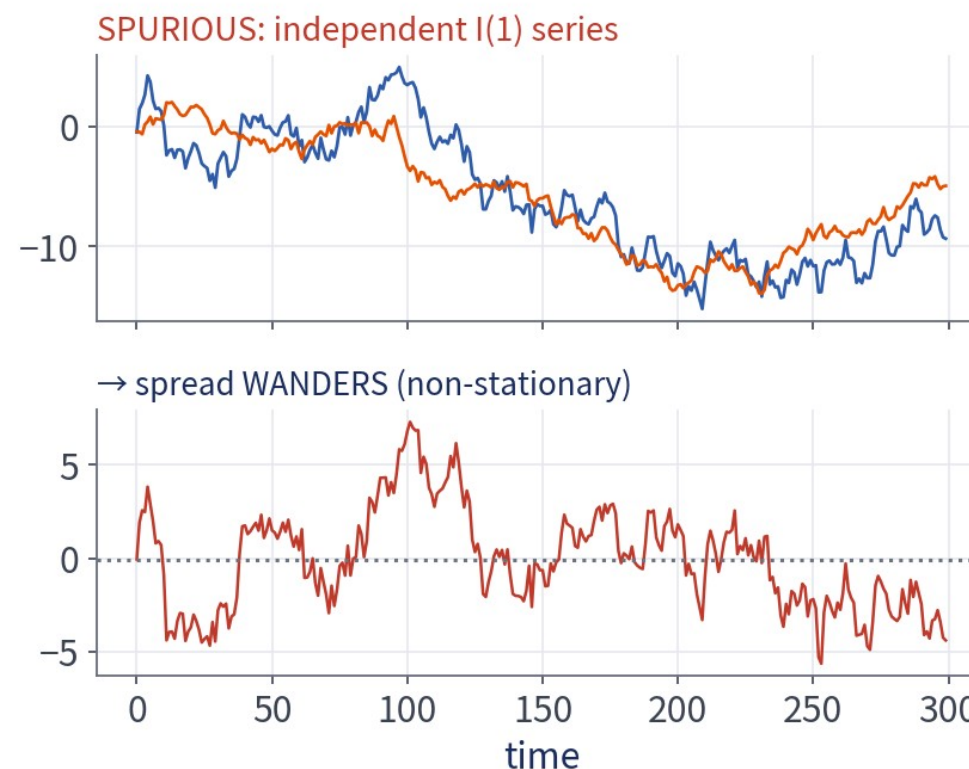
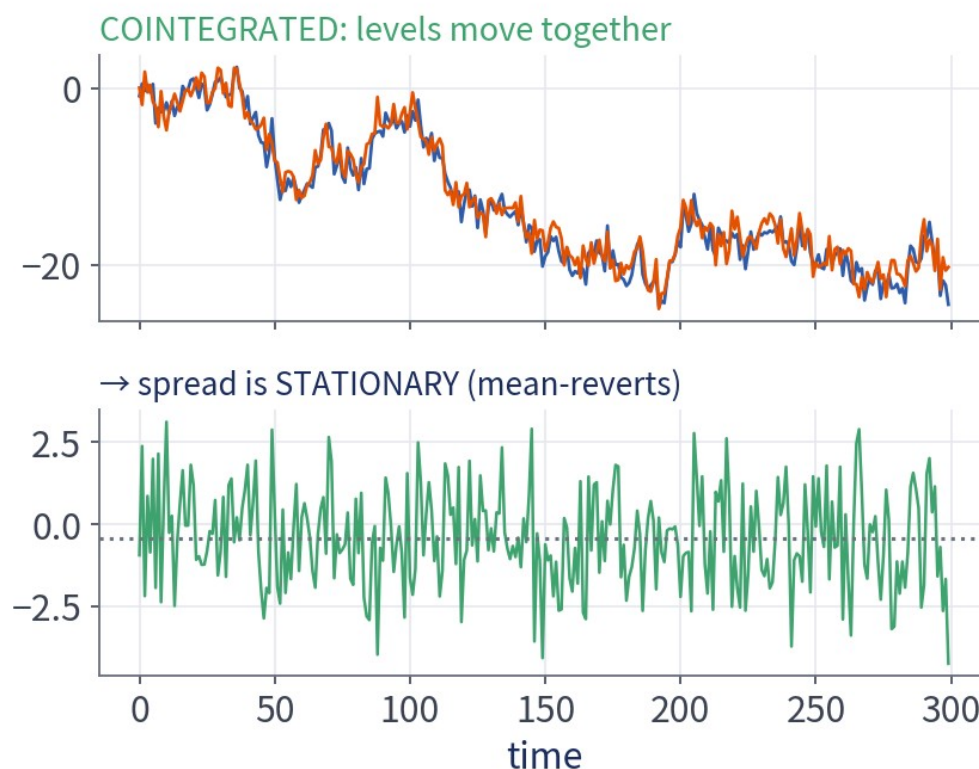
What it guarantees

- The spread has a fixed mean it keeps returning to.
- Deviations are temporary and self-correcting.
- So the levels carry real, usable long-run information.

Cointegrated vs Spurious — Look at the Spread

The single distinguishing test: if the spread between the levels is stationary (left), they are cointegrated; if it wanders (right), the regression is spurious.

Cointegration is what separates a real long-run link from a spurious one



The Engle-Granger Two-Step Test

Test cointegration with tools you already have

$$Y_t = \alpha + \beta X_t + Z_t$$

step 1: regress levels

ADF test on $\hat{Z}_t$

step 2: is it $I(0)$?

Step 1: regress the levels

- Regress Y on X and keep the residual, the estimated spread.
- This $\hat{\beta}$ is your candidate cointegrating coefficient.
- Ignore its t-statistic here — that is not the test.

Step 2: test the residual

- Run an ADF-type test on the residual.
- If the residual is $I(0)$, the series ARE cointegrated.
- If it is $I(1)$, the regression was spurious after all.

The Error Correction Model (ECM)

How the spread pulls the series back

$$\Delta Y_t = c + \gamma \underbrace{Z_{t-1}}_{\text{last error}} + \sum_j \phi_j \Delta X_{t-j} + \varepsilon_t, \quad \gamma < 0$$

The two forces

- Short-run: changes in X move Y period to period.
- Long-run: the last spread Z_{t-1} pulls Y back toward equilibrium.
- $\gamma < 0$ is the speed of that correction.

Why it is elegant

- It uses the levels (via the spread) AND stays valid.
- It unifies short-run dynamics with the long-run tether.
- It is the natural model once cointegration is confirmed.

The Johansen Test

When more than two series share equilibria

Engle-Granger handles a single pair. The Johansen test generalizes to many series at once and can find MORE than one cointegrating relationship among them.

What it adds

- A system approach for several $I(1)$ series together.
- It counts how many independent equilibria exist.
- The standard tool for baskets, curves, and multi-asset spreads.

When you need it

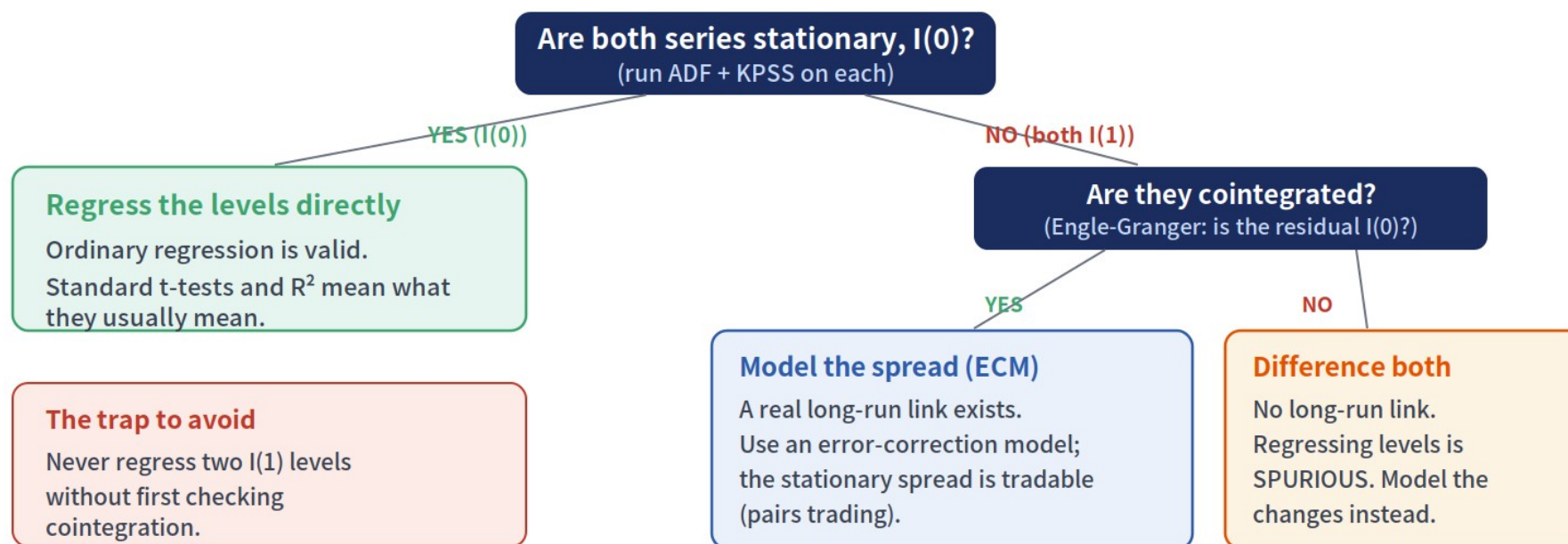
- Yield curves — many maturities tied together.
- A basket of related stocks or commodities.
- Any time a single pair is too narrow a lens.

Two Series, One Decision

The whole of Week 2 as a single flow: test for $I(1)$, then for cointegration, and let the answer choose your model — levels, the spread, or differences.

Two Series, One Decision: How to Regress Them Safely

The whole of Week 2, as a single flow. Follow it before you ever regress two levels.



Pairs Trading: Cointegration at Work

The strategy the theory makes possible

If two assets are cointegrated, their spread mean-reverts. Sell the spread when it is stretched wide, buy it when it is compressed, and profit as it returns to its mean.

The recipe

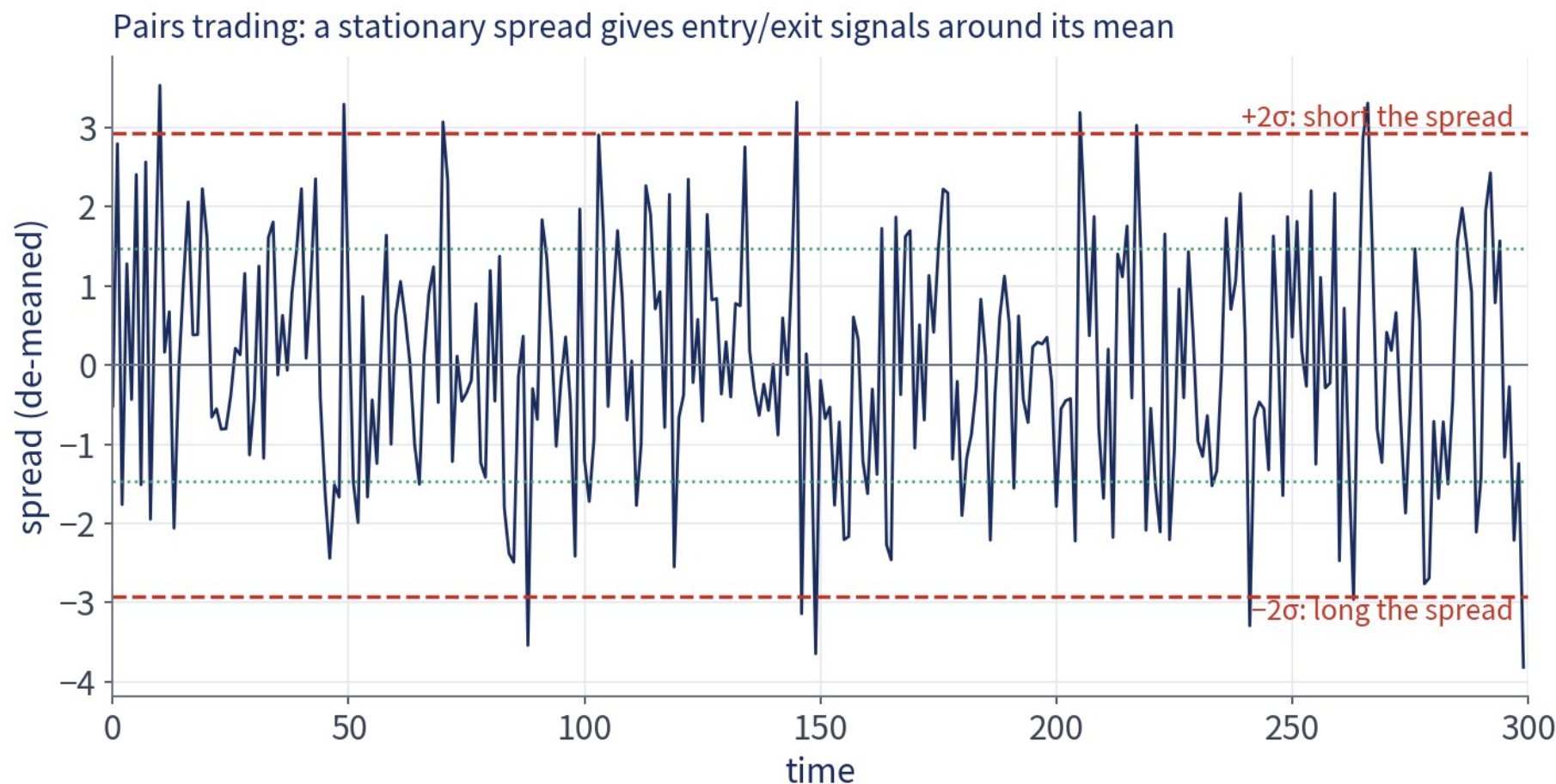
- Find a cointegrated pair; estimate the spread and its mean.
- Enter when the spread deviates far (e.g. ± 2 standard deviations).
- Exit as it reverts back toward the mean.

Why it can work

- The trade is market-neutral — it bets on the tether, not direction.
- Cointegration is the statistical license for 'reversion to a mean'.
- Recall Part 1: reversion needs a stationary series — here, the spread.

The Tradable Spread

A stationary spread oscillates around its mean. The dashed bands mark where it is 'stretched' — the entry signals for a mean-reversion (pairs) trade.



Cointegration Can Break

The tether is not guaranteed to hold

Where it fails

- A structural break — a merger, a regime change — can snap the link.
- A relationship estimated in-sample may not persist out-of-sample.
- Transaction costs can erase a small, slow-reverting edge.

How to stay safe

- Re-test the relationship regularly; do not assume it is permanent.
- Validate out-of-sample before committing capital.
- Size positions for the day the spread does NOT revert.

The Two-Series Workflow

From raw levels to a valid model

1 • Test $I(1)$

each series

Confirm both are individually non-stationary (Part 2).

2 • Regress

get the spread

Regress the levels; keep the residual as the spread.

3 • Test spread

cointegration

ADF-test the residual: is it $I(0)$?

4 • Model

ECM or diff

If cointegrated, build an ECM; if not, difference.

5 • Monitor

stay honest

Re-test over time; the tether can break.

This flow turns the spurious-regression trap into a disciplined, testable routine.

Reproducing 1974, then Finding a Real Pair

Direct Claude Code — see the trap and the cure yourself

GOAL Reproduce the spurious-regression result, then test two real assets for cointegration and build the spread.

[INPUT] Prompt · Claude Code

Write and run `cointegration_lab.py` that:

1. simulates 1000 pairs of INDEPENDENT random walks, regresses each pair, and reports how often $|t| > 1.96$ (the spurious rate);
 2. downloads two related tickers (e.g. KO and PEP), tests each for $I(1)$, then runs the Engle-Granger cointegration test on them;
 3. if cointegrated, plots the spread with its mean and $\pm 2\sigma$ bands and marks entry signals.
- Explain, in plain English, what each result means.

How to work with it

- Expect the spurious rate near 70–90%, not 5%.
- Check the residual's ADF p-value for the pair.
- Try an UNrelated pair — cointegration should fail.
- This ties Parts 1–3 into one workflow.

Deliverable: `cointegration_lab.py` + a spread chart for your chosen pair.

Three Questions Before You Move On

If you can answer these, Part 3 has landed

Q1

You regress two $I(1)$ prices and get $R^2 = 0.9$, huge t-stat. Believe it?

No — test residual

Q2

Both series are $I(1)$ and their spread is $I(0)$. What can you build?

An ECM / pairs trade

Q3

Why is differencing a valid but sometimes wasteful fix?

Loses the long run

All three hinge on one question: what is the residual doing?

Three Pitfalls to Avoid

Where the cure gets misapplied

Common mistakes

- "A high R^2 between two prices proves a real relationship."
- "Cointegration, once found, lasts forever."
- "The Engle-Granger t-stat in step 1 is the test."

The reality

- R^2 on $I(1)$ levels is meaningless — only the residual's order matters.
- Relationships break; re-test and validate out-of-sample.
- The test is the ADF on the residual (step 2), not step 1's t-stat.

Week 2 in Three Ideas

Pulling the whole week together

1

Stationarity is the gatekeeper

Part 1: model only stable statistics. Weak stationarity, the ACF, and the AR(1) lens tell a mean-reverting series from a wandering one.

2

Unit roots are detected and cured by testing

Part 2: ADF and KPSS together decide; a unit root is differenced, a deterministic trend is detrended — never the reverse.

3

Cointegration separates real links from mirages

Part 3: unrelated $I(1)$ series fake relationships; a stationary spread reveals the genuine ones — the basis of the error-correction model and pairs trading.

Next — Week 3: ARMA Models, building forecasts on the stationary series you now know how to make.